

Arm Take-home Assignment (AgileX Piper / Similar Manipulator)

Section 1: Kinematics & Reachability

Objective

Evaluate candidate's ability to model the arm, solve IK, and reason about workspace constraints.

Setup

• Use MoveIt 2 or equivalent with a 6-DOF arm model (Piper or similar). • Scene includes: countertop slab (120×60 cm), faucet obstacle, vertical mirror (90×60 cm).

Tasks

- 1 Configure arm + planning scene with collision objects.
- 2 Implement an IK service that solves for reachable end-effector poses (surface-aligned) and rejects infeasible solutions.
- 3 Generate a reachability heatmap for a 60×60 cm patch of counter at 2 cm resolution.

Deliverables

- 1 URDF/config + IK solver node.
- 2 Reachability CSV + visualization (heatmap).
- 3 Short write-up: "Where can/can't the arm reach, and why?".

Section 2: Surface Coverage Path Planning

Objective

Assess ability to generate efficient, constraint-aware toolpaths for wiping.

Setup

• Tool: rectangular pad, 100×50 mm footprint. • Surfaces: countertop + mirror. • Constraints: Tool normal $\approx$ surface normal $\pm 10^\circ$, keep-out margin 15 mm, desired overlap 10–20%.

Tasks

- 1 Implement at least one coverage strategy (e.g., raster). Bonus: implement spiral for mirror.
- 2 Convert Cartesian waypoints $\rightarrow$ joint trajectories with time parameterization.
- 3 Report metrics: coverage %, path length, estimated execution time.

Deliverables

- 1 Planner node + visualization (RViz or plots).
- 2 Example joint trajectory output.
- 3 Short comparison of strategies (raster vs spiral).

Section 3: Contact-Aware Wiping Control

Objective

Test candidate's understanding of force/impedance control in contact tasks.

Setup

• Simulated wrist force/torque sensor. • Targets: Countertop: 10 N $\pm$ 2 N, 0.15–0.25 m/s. Mirror: 6 N $\pm$ 1.5 N, 0.1–0.2 m/s.

Tasks

- 1 Implement controller that switches to force control when $|F_z| > 2$ N.
- 2 Maintain target force while following the path, and back off if $|F_z| > 15$ N.
- 3 Show handling of an obstacle (e.g., faucet) by skipping or replanning locally.
- 4 Log and plot force vs time and velocity vs time.

Deliverables

- 1 Controller code + configs.
- 2 Plots of force/velocity tracking.
- 3 Short demo (video/gif or sim run).

General Expectations

- 1 Provide clear build/run instructions.
- 2 Keep configs (forces, overlaps, tool size) parameterized.
- 3 Include short README with design notes and trade-offs.